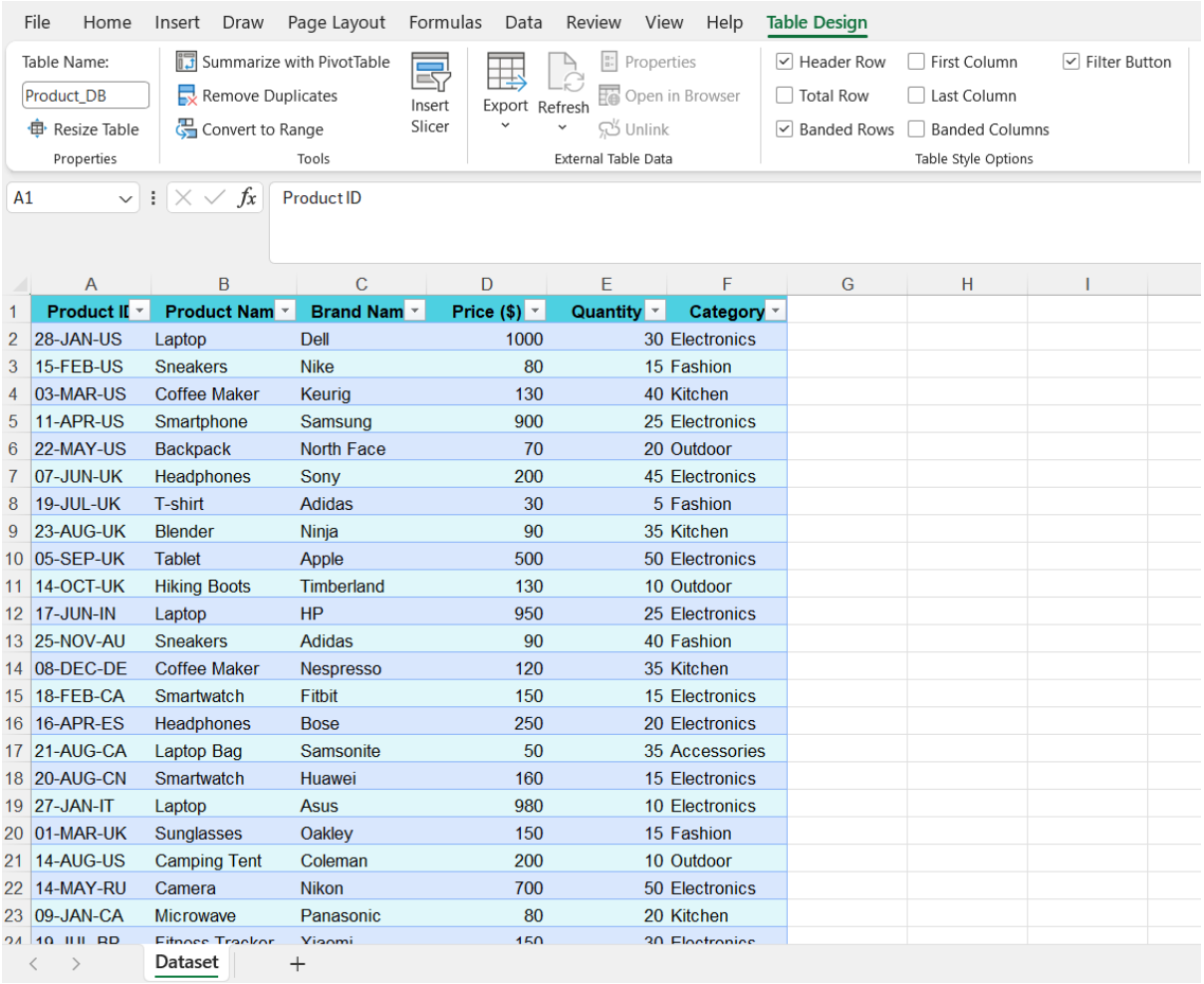
Data Analytics (DA)

Module -1 - Excel

Excel Assignment 1 – Data Exploration

Tasks:-

Creating a Table Name as **Product_DB**

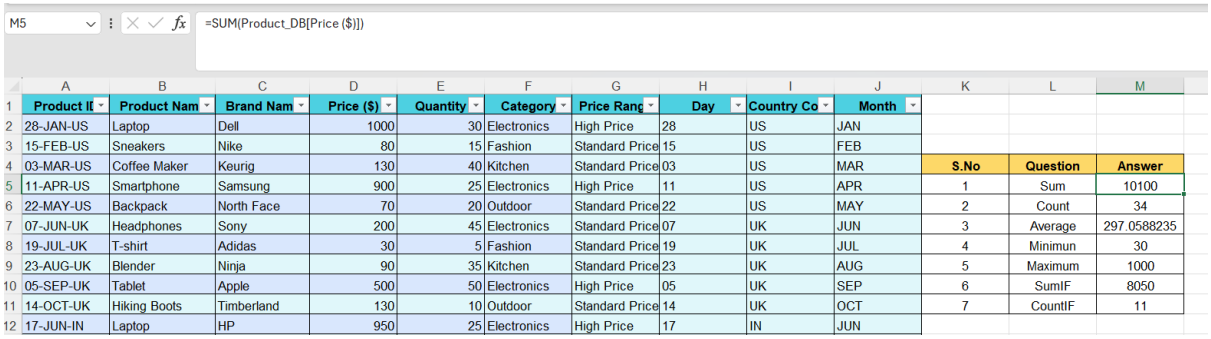


	Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category
2	28-JAN-US	Laptop	Dell	1000	30	Electronics
3	15-FEB-US	Sneakers	Nike	80	15	Fashion
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
5	11-APR-US	Smartphone	Samsung	900	25	Electronics
6	22-MAY-US	Backpack	North Face	70	20	Outdoor
7	07-JUN-UK	Headphones	Sony	200	45	Electronics
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen
10	05-SEP-UK	Tablet	Apple	500	50	Electronics
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
12	17-JUN-IN	Laptop	HP	950	25	Electronics
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16	16-APR-ES	Headphones	Bose	250	20	Electronics
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
19	27-JAN-IT	Laptop	Asus	980	10	Electronics
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
22	14-MAY-RU	Camera	Nikon	700	50	Electronics
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen

Dataset

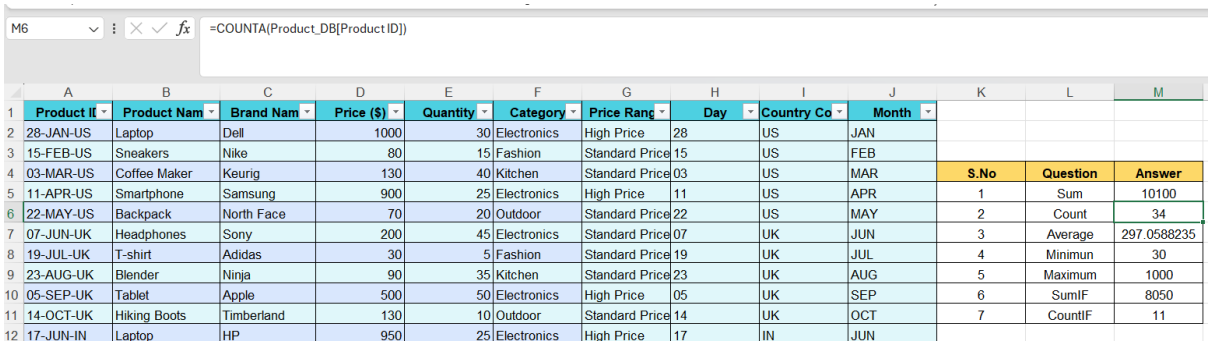
1. **Sum, Count, and Average:-**
   - Sum – Calculate the total price of the all the Products
   - Count – Calculating the count of products exists in the dataset
   - Average – Calculating average price of the product.

S.No	Question	Answer
1	Sum	10100
2	Count	34
3	Average	297.0588235



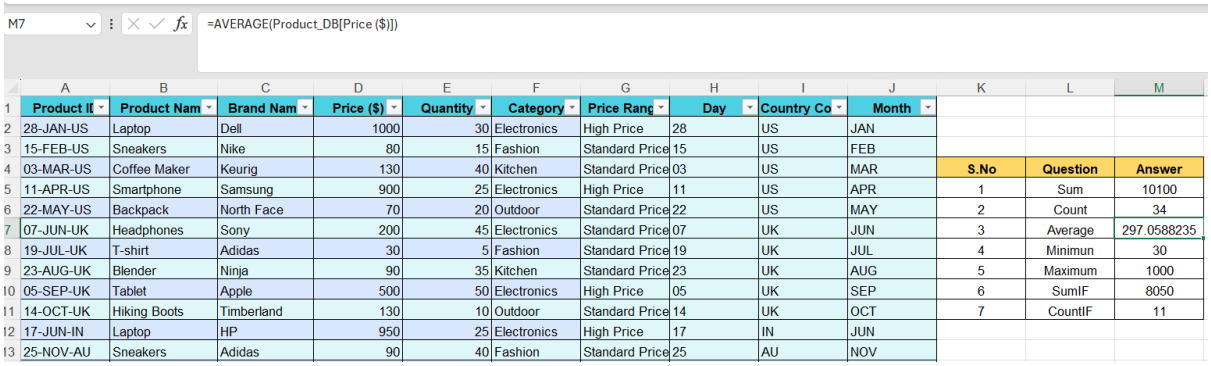
M5 =SUM(Product_DB[Price ($)])

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Product II	Product Nam	Brand Nam	Price ($)	Quantity	Category	Price Rang	Day	Country Co	Month			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN			
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB			
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR	S.No	Question	Answer
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR	1	Sum	10100
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY	2	Count	34
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN	3	Average	297.0588235
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL	4	Minimun	30
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG	5	Maximum	1000
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP	6	SumIF	8050
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT	7	CountIF	11
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN			



M6 =COUNTA(Product_DB[Product ID])

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Product II	Product Nam	Brand Nam	Price ($)	Quantity	Category	Price Rang	Day	Country Co	Month			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN			
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB			
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR	S.No	Question	Answer
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR	1	Sum	10100
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY	2	Count	34
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN	3	Average	297.0588235
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL	4	Minimun	30
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG	5	Maximum	1000
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP	6	SumIF	8050
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT	7	CountIF	11
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN			



M7 =AVERAGE(Product_DB[Price ($)])

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Product II	Product Nam	Brand Nam	Price ($)	Quantity	Category	Price Rang	Day	Country Co	Month			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN			
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB			
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR	S.No	Question	Answer
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR	1	Sum	10100
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY	2	Count	34
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN	3	Average	297.0588235
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL	4	Minimun	30
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG	5	Maximum	1000
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP	6	SumIF	8050
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT	7	CountIF	11
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN			
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV			

2. Minimum and Maximum Values:-

- Minimum – Finding the minimum price of the product
- Maximum - Finding the maximum price of the product

S.No	Question	Answer
4	Minimun	30
5	Maximum	1000

M8 =MIN(Product_DB[Price ($)])

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV

S.No	Question	Answer
1	Sum	10100
2	Count	34
3	Average	297.0588235
4	Minimun	30
5	Maximum	1000
6	SumIF	8050
7	CountIF	11

M9 =MAX(Product_DB[Price ($)])

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV

S.No	Question	Answer
1	Sum	10100
2	Count	34
3	Average	297.0588235
4	Minimun	30
5	Maximum	1000
6	SumIF	8050
7	CountIF	11

3. Logical Function – IF:-

- Crating a new column as Price Range and,
- Using IF function classified Price>=500 as Highprice and Price<500 as Standard Price

G2 =IF(D2>=500,"High Price","Standard Price")

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN

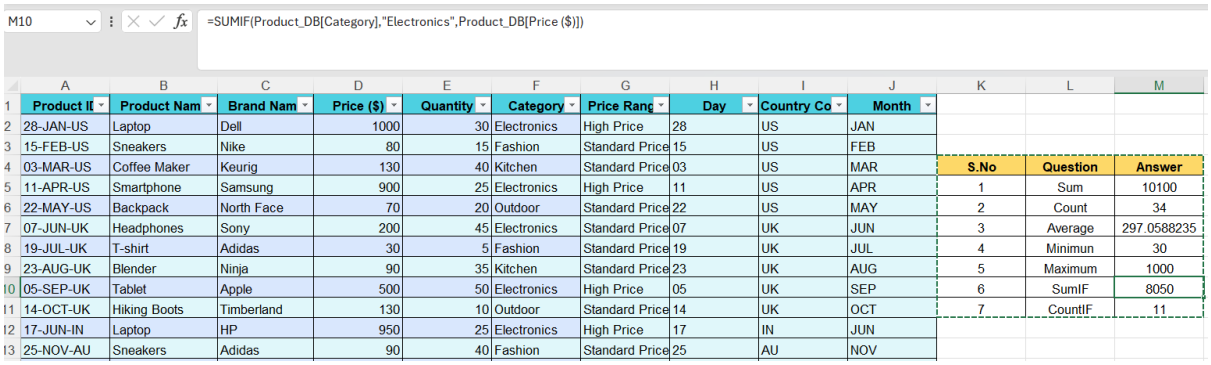
S.No	Question	Answer
1	Sum	10100
2	Count	34
3	Average	297.0588235
4	Minimun	30
5	Maximum	1000
6	SumIF	8050
7	CountIF	11

4. Conditional Functions – SUMIF and COUNTIF

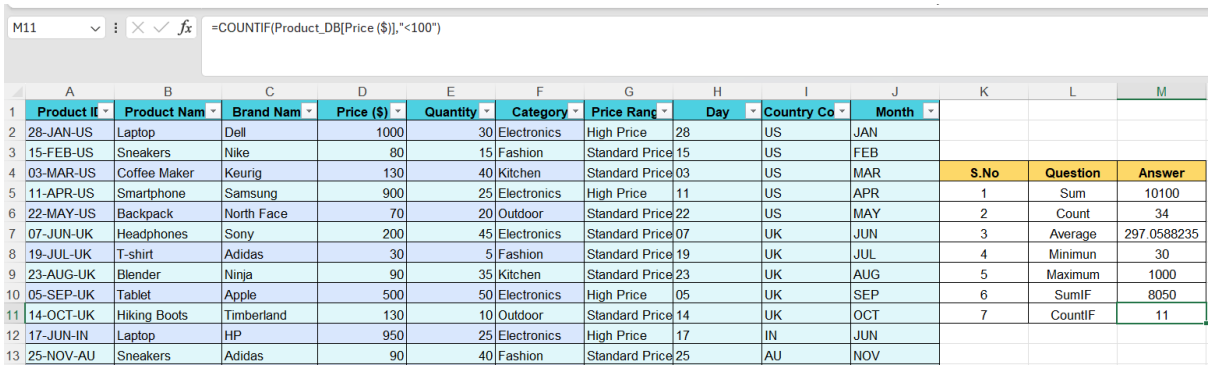
- SUMIF - calculating the total price of products in the Electronics category.

- COUNTIF - determining the number of products with a price less than $100.

S.No	Question	Answer
6	SumIF	8050
7	CountIF	11



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category	Price Rang	Day	Country Co	Month			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN			
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB			
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR	S.No	Question	Answer
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR	1	Sum	10100
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY	2	Count	34
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN	3	Average	297.0588235
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL	4	Minimun	30
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG	5	Maximum	1000
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP	6	SumIF	8050
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT	7	CountIF	11
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN			
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV			



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category	Price Rang	Day	Country Co	Month			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN			
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB			
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR	S.No	Question	Answer
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR	1	Sum	10100
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY	2	Count	34
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN	3	Average	297.0588235
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL	4	Minimun	30
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG	5	Maximum	1000
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP	6	SumIF	8050
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT	7	CountIF	11
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN			
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV			

5. Text Formatting – LEFT, RIGHT, MID:-

1. Create a new column named Day with the first 2 characters of each 'Product ID' using the LEFT function.
2. Create a new column named Country Code by extracting the last 2 characters from the 'Product ID' column using the RIGHT function.
3. Create a new column named Month by extracting the 4th to 6th characters from the 'Product ID' column using the MID function.

H2 =LEFT(A2,2)

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category	Price Rang	Day	Country Co	Month
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV

I2 =RIGHT(A2,2)

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category	Price Rang	Day	Country Co	Month
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC

J2 =MID(A2,4,3)

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Nam	Brand Nam	Price ($)	Quantity	Category	Price Rang	Day	Country Co	Month
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB